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RESEARCH ARTICLE

Improving Handwritten Mathematical Expression Recognition via Integrating Convolutional Neural Network With Transformer and Diffusion-Based Data Augmentation

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ABSTRACT Handwritten mathematical expression recognition (HMER) poses a formidable challenge due to the intricate two-dimensional structures and diverse handwriting styles. This paper introduces a novel approach to improve HMER accuracy by employing an integrated, high-capacity architecture that combines Transformer and Convolutional Neural Network (CNN) models, along with a denoising diffusion probabilistic model (DDPM)-based data augmentation technique. We explore three combination strategies for an attention-based encoder-decoder (AED) HMER model: 1) The ''Tandem'' strategy, which harnesses CNN features within a Transformer encoder to capture global interdependencies; 2) The ''Parallel'' strategy, which integrates Transformer encoder outputs with CNN outputs to achieve comprehensive feature fusion; 3) The ''Mixing'' strategy, which introduces multi-head self-attention (MHSA) at the final stage of the CNN. We evaluate these methods using the CROHME benchmark dataset and conduct a detailed comparative analysis. All three approaches significantly enhance model performance. Notably, the ''Tandem'' approach achieves expression recognition rates (ExpRate) of 54.85% and 58.56% on the CROHME 2016 and 2019 test sets, respectively, while the ''Parallel'' method attains 55.63% and 57.39% on the same test sets. Furthermore, we introduce an innovative data augmentation approach that utilizes DDPM to generate synthetic training samples. The DDPM, conditioned on LaTeX-rendered images, bridges the gap between printed and handwritten expressions, enabling the creation of realistic, stylistically diverse handwriting samples. This augmentation boosts the ExpRates of all strategies on both CROHME 2016 and 2019 test sets, yielding improvements of 1.6-4.6% relative to the unaugmented dataset.

INDEX TERMS CNN, data augmentation, denoising diffusion probabilistic model, DDPM, handwritten mathematical expression recognition, HMER, Transformer.

I. INTRODUCTION

Handwritten Mathematical Expression Recognition (HMER) holds significant importance in various applications, including intelligent office and education systems. It simplifies the arduous task of manually inputting handwritten formulas by converting them into LaTeX text, facilitating computer compilation and editing. However, HMER presents unique challenges compared to traditional Optical Character Recognition (OCR) due to its stringent accuracy requirements for symbol type, position, and the spatial relationships between symbols [1]. Consequently, designing an effective HMER

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system is a complex endeavor, necessitating abundant and representative training data.

The advent of deep learning has led to the adoption of attention-based encoder-decoder (AED) models in HMER solutions [5], [6], [7], [46]. For instance, the ''Watch, attend and parse'' (WAP) model [46] introduced an end-to-end neural network-based approach, utilizing a Convolutional Neural Network (CNN) as the encoder and a Recurrent Neural Network (RNN) as the decoder to generate LaTeX-formatted recognition results. This pioneering work achieved much better recognition accuracy over traditional methods. Subsequently, the AED model has gained prominence as the mainstream solution for HMER, inspiring various innovative approaches aimed at enhancing its performance from diverse perspectives.

Reference [44] introduced a multi-scale attention mechanism within the WAP framework, enhancing symbol recognition across different scales while recovering fine-grained details often lost during pooling layers. References [40] and [41] pioneered paired adversarial learning, aiding the model in prioritizing semantic-invariant features, particularly valuable in addressing challenges stemming from writing-style variations and limited dataset sizes. Reference [19] proposed pattern generation strategies, offering a novel approach to improving HMER accuracy. Reference [17] incorporated a transition probability matrix into the decoder, facilitating the capture of long-range dependencies and fostering syntactical rule learning. Reference [45] explored the replacement of WAP's string decoder with a tree-structured decoder, explicitly modeling mathematical expression trees for enhanced structural understanding. Reference [20] employed scale augmentation and drop attention techniques, effectively elevating HMER accuracy. Reference [36] introduced an auxiliary symbol recognition loss, providing weak supervision signals to enhance HMER accuracy. Reference [29], a global context block was integrated into both the CNN and Transformer encoders, augmenting their capacity to encode positional information. Reference [42] improved efficiency by replacing the RNN-based decoder in a printed math expression recognition system with a CNN-based alternative. Reference [49] elevated accuracy by introducing multi-head attention and a stacked decoder design into the WAP framework. Reference [30] combined graph neural networks with CNN and RNN encoders, enhancing the modeling of 2D spatial structures within math expression images. Reference [48] introduced the Bidirectionally Trained Transformer (BTTR) model, streamlining attention map generation for improved accuracy. Reference [43] integrated syntax information into the AED framework, offering promising enhancements to HMER systems. Reference [50] leveraged attention aggregation module and bidirectional mutual learning to improve recognition accuracy. In [51], a counting aware network was introduced to improve model performance on duplicated symbol patterns. Reference [52] used semantic graph representation learning to enhance the modeling capacity of HMER system and [53] conducted a thorough investigation on model architecture design including ralative positional embedding, bidirectional decoding, siamese network.

Recently, there has been a surge of interest in Transformer [38] architectures due to their performance in various domains, such as natural language processing (NLP) [8], and computer vision (CV) [2], [11]. Consequently, researchers have embarked on efforts to combine CNNs and Transformers, aiming to harness the strengths of both paradigms for enhanced model performance. Notable examples of this fusion include BoTNet [35], CoAtNet [4], and Conformer [13]. BoTNet, building upon the foundation of ResNet [14], replaces the traditional 3×3 convolutional kernel with Multi-Head Self Attention (MHSA) modules in the final stage of the CNN bottleneck. This architectural tweak has yielded improvements in various tasks, including instance segmentation and object detection, all while reducing the model's parameter size. CoAtNet takes an approach by seamlessly uniting convolutional and self-attention modules through relative attention mechanisms and vertically stacking these layers. This hybridization empowers CoAtNet to mitigate Transformer's inherent inductive bias and enhance its adaptability in resource-constrained settings, resulting in improved generalization and efficiency. In speech area, Conformer represents a compelling fusion of CNN and Transformer elements. Within its architecture, Conformer integrates a modified encoder consisting of six blocks, each incorporating both feed-forward network (FFN) and MHSA modules, complemented by Convolutional layers. By blending the local feature extraction capabilities of CNNs with the global dependency modeling capacity of Transformers, Conformer presents a potent framework that advances the state of the art in speech-related tasks.

In the quest to synergize CNNs and Transformers for optimal performance, this paper takes a unique approach by employing a Transformer decoder and focusing on the structural modifications within the encoder. The study introduces three novel combinations:

1) *Tandem* approach: Leveraging a CNN encoder, specifically DenseNet, to extract local features from handwritten mathematical expression images. These local features are then input into the Transformer encoder to capture global dependencies. This method effectively combines the strengths of local feature extraction, global dependency modeling, and interaction information integration, thereby enhancing overall model performance.
2) *Parallel* approach: Simultaneously processing raw image patches through both CNN and Transformer dual-branch encoders. The outputs of these two branches are then concatenated to yield the final encoder output. This approach capitalizes on the parallel processing capabilities of both CNN and Transformer, enabling comprehensive feature extraction and information fusion.

3) *Mixing* approach: Replacing the convolutional layer in the final stage of the CNN encoder with a MHSA module. This modification facilitates the capture of global dependencies while maintaining a streamlined parameter count, effectively harnessing the advantages of the Transformer model.

Across the CROHME2014 [25], 2016 [26], and 2019 [24] datasets, the performance of the *Tandem* and *Parallel* approaches exhibits minimal disparity, with both outperforming the *Mixing* approach. Notably, the *Parallel* approach marginally outperforms the *Tandem* approach in terms of recognition accuracy.

To maximize the capabilities of these model architectures and further enhance HMER accuracy, we also explored data augmentation in HMER using latest generative modeling techniques. Recently, denoising diffusion probabilistic models (DDPMs) have been emerged as state-of-the-art in various image generation tasks. These tasks span from text-to-image generation [9], [32], [33], [34] to image-to-image generation [22], [39]. In the context of generating handwritten images, [23] explored the application of a text-to-image DDPM for online handwritten content generation and yielded promising outcomes. Likewise, [12] introduced a glyph conditional DDPM (GC-DDPM) for the generation of offline handwritten Chinese characters and handwritten English text lines [10].

Drawing inspiration from the accuracy enhancements achieved by these techniques, our approach involves the synthesis of handwritten mathematical expression samples using GC-DDPM. We then integrate this synthetic data with real data to construct an HMER system, with the aim of further elevating its performance levels. Experimental results show that DDPM-based augmentation further boosts the ExpRates of proposed architectures by 1.6-4.6% in relative.

Our main contributions include:

1) A thorough exploration of combining CNNs and Transformers for HMER tasks: Our study introduces three novel architectures—Tandem, Parallel, and Mixing approaches. These architectures combine the strengths of CNNs and Transformers to enhance the recognition accuracy by leveraging local and global contextual features.
2) Use of DDPMs for HMER data augmentation: We explore the use of DDPMs for data augmentation, which helps in generating synthetic images of mathematical expressions. This approach not only enhances the diversity of the training dataset but also improves the model's ability to generalize from synthetic to real handwritten mathematical expression data. To the best of our knowledge, this is the first effort to improve HMER performance by leveraging DDPM-synthesized training samples.
3) Combining proposed architectures and DDPM-generated data, we achieved state-of-the-art results on benchmark datasets, which demonstrating the effectiveness of our proposed approaches.

It's worth noting that this paper represents an extension of our prior work [47], delving deeper into DDPM-based data augmentation. The structure of the remainder of this paper unfolds as follows: Section II elaborates on our exploration of model architecture, followed by an introduction to GC-DDPM in Section III. Section IV presents a detailed account of our experimental results, and, finally, Section V serves as a conclusion to this work.

II. MODEL ARCHITECTURE DESIGN

A. BASELINE SYSTEM

Our study employs the BTTR [48] as the foundational architecture for our baseline system, depicted in Figure 1. The BTTR eschews the conventional RNN-based decoding process in favor of a Transformer-based decoder, which enhances the system's ability to generate concise output. In the encoding phase, we utilize the DenseNet architecture [18], renowned for its dense connectivity pattern, to facilitate feature extraction. DenseNet's architecture, by integrating direct connections from each layer to all subsequent layers, ensures an efficient flow of information, mitigates the vanishing gradient problem, and reduces the total number of parameters.

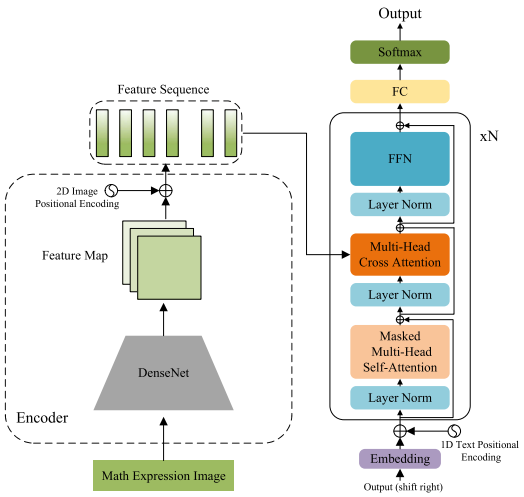


FIGURE 1. Overview of our baseline model.

For the decoding phase, our approach leverages the Transformer decoder's bidirectional training capability. This design allows for the input LaTeX sequence to be presented in both original and reverse order, tagged with start-of-sequence (< SOS >) and end-of-sequence (< EOS >) tokens. The bidirectional nature of the Transformer means it treats input sequences as a set rather than a list, allowing it to process forward and backward sequences concurrently during training. This sequence-agnostic processing is instrumental in maintaining the integrity of the output without compromising the accuracy of recognition.

The addition of a two-dimensional absolute positional encoding post feature extraction via DenseNet is crucial. It enriches the Transformer decoder with spatial coordinates for each image feature vector, thereby enhancing its ability to

pinpoint and interpret the location-specific attributes of the input data.

In this work, we adopt the BTTR model as our primary framework, introducing two enhancements to refine the system's performance. First, we substitute the ReLU activation function in the FFN with the Gaussian Error Linear Unit (GELU) [15]. The GELU activation function is known to synergize well with Transformer architectures, offering benefits such as accelerated learning, alleviation of vanishing gradient issues, and provision of richer statistical significance due to its non-sparse output compared to ReLU.

Second, we revise the decoder's normalization technique, transitioning from post-normalization to pre-normalization. By positioning the layer normalization before both the MHSA and FFN within the decoder, the model's sensitivity to initial parameter selection is reduced, thus streamlining the training process by obviating the need for an extensive parameter-tuning phase and eliminating the necessity for warm-up training procedures.

The methodology proposed in this paper draws inspiration from recent advancements in textual document recognition, specifically the approach outlined by [3], which employs a fully convolutional network (FCN) encoder and a Transformer decoder for semantic-level image segmentation to enhance the accuracy of document recognition. This comparative analysis serves as a reference for our proposed modifications, underscoring our commitment to elevating the standard of HMER.

B. TANDEM APPROACH

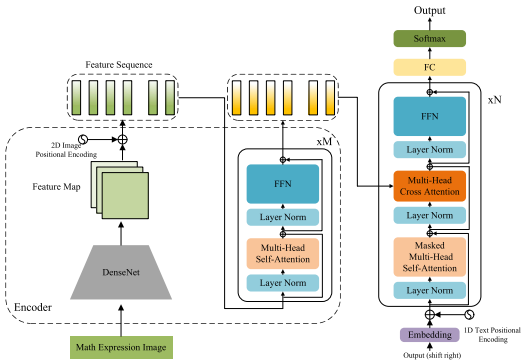


FIGURE 2. Overview of our *Tandem* approach.

Building on the promising outcomes achieved by incorporating a Transformer decoder into the BTTR model, we extend our architectural enhancements to the encoder segment of our baseline model by integrating a Transformer encoder. This modification facilitates a tandem connection with the pre-existing CNN encoder, as illustrated in Figure 2. We refer to this novel configuration as the "Tandem" approach. In this setup, the feature maps produced by the CNN encoder are fed directly into the Transformer encoder. By processing through multiple layers of the Transformer encoder, the resulting output is enriched with a broader context of global information. It is our hypothesis that this enriched global contextual understanding will translate into improved accuracy in mathematical expression recognition.

C. PARALLEL APPROACH

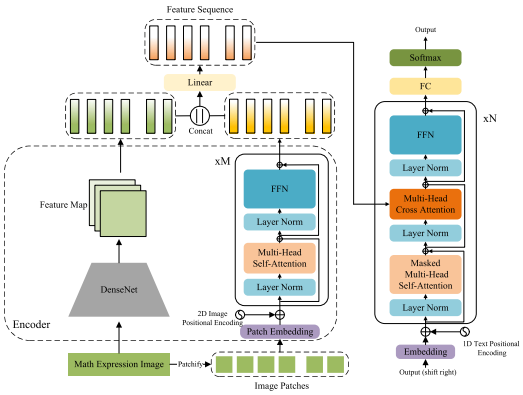


FIGURE 3. Overview of our *Parallel* approach.

In line with the principles of the Tandem approach, our methodology also seeks to combine both CNN and Transformer encoders in parallel. Initially, the features extracted from the original image data undergo simultaneous processing by these two encoders. Subsequently, the resulting two feature sequences are concatenated, generating the final feature sequence through a linear mapping layer, as illustrated in Figure 3.

Within the Transformer branch, we adopt a procedure similar to that of ViT [11], where the input image is divided into a sequence of 16×16 patches. This sequence is then passed through a linear embedding layer, creating the input sequences for the Transformer encoder.

Our intention is for the CNN and Transformer encoders to independently capture both local and global contextual information, ultimately enhancing the overall recognition accuracy.

D. MIXING APPROACH

Taking inspiration from BoTNet [35], we have explored the substitution of 3×3 convolution layers in the final stage of DenseNet with MHSA layers. This innovative approach, which we refer to as the "Mixing" approach, leads to the creation of a novel model termed "DenseSANet". The key distinction between DenseNet and DenseSANet lies in the integration of MHSA within the final stage of the latter.

A detailed architectural comparison between DenseNet and DenseSANet can be found in Table 1. In addition to replacing the 3×3 convolution layers with MHSA, we have also made further adjustments by substituting Batch Normalization (BN) with Instance Normalization (IN) [37] and replacing the ReLU activation function with GELU.

In contrast to the "Tandem" and "Parallel" approaches, we anticipate that the "Mixing" approach offers a more parameter-efficient means of capturing both local and global contextual information.

TABLE 1. Architecture comparison of DenseNet and DenseSANet. Assume input image size as 256 × 256.

Layers	Output Size	DenseNet	DenseSANet
Convolution	128 × 128	7 × 7 conv+BN+ReLU, stride 2	
Pooling	64 × 64	2 × 2 max pool, stride 2	
Dense (1)	64 × 64	$\begin{bmatrix} 1\times1 \text{ conv+BN+ReLU} \\ 3\times3 \text{ conv+BN+ReLU} \end{bmatrix} \times 16$	
Transition (1)	64 × 64	1 × 1 conv+BN+ReLU	
	32 × 32	2 × 2 average pool, stride 2	
Dense (2)	32 × 32	$\begin{bmatrix} 1\times1 \text{ conv+BN+ReLU} \\ 3\times3 \text{ conv+BN+ReLU} \end{bmatrix} \times 16$	
Transition (2)	16 × 16	1 × 1 conv+BN+ReLU	
	32 × 32	2 × 2 average pool, stride 2	
Dense (3)	16 × 16	$\begin{bmatrix} 1\times1 \text{ conv+BN+ReLU} \\ 3\times3 \text{ conv+BN+ReLU} \end{bmatrix} \times 16$	$\begin{bmatrix} 1\times1 \text{ conv+IN+GELU} \\ \text{MHSA} \\ 1\times1 \text{ conv+IN+GELU} \end{bmatrix} \times 16$
Post Norm	16 × 16	BN	

III. GLYPH CONDITIONAL DDPM

Building upon the GC-DDPM methodology introduced by [12], our approach leverages a GC-DDPM for the generation of handwritten images that faithfully represent the content of a provided LaTeX math string. In our process, each input LaTeX math string is rendered into a printed glyph image through the TeX rendering engine. Opting for a glyph image as our input holds distinct advantages, as it contains the information of symbol shapes and intricate layouts, surpassing the informational depth of a plain LaTeX string. This glyph image is denoted as $\mathbf{g}$.

As illustrated in Figure 4, let $\mathbf{x}_0$ represent a data point sampled from a real distribution, i.e., $\mathbf{x}_0 \sim q(\mathbf{x})$, which is associated with a corresponding glyph image $\mathbf{g}$. In the forward diffusion process, we introduce small amounts of Gaussian noise to $\mathbf{x}$ over T steps, resulting in a sequence $\mathbf{x}_{t=1}^{T}$, which is computed as follows:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}), \tag{1}$$

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\epsilon_t \tag{2}$$

where $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\beta_t \in (0, 1)$, and $\alpha_t = 1 - \beta_t$. It is worth noting that:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \tag{3}$$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \tag{4}$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\bar{\alpha}_t = \prod_{i=1}^{t}\alpha_i$. In the limit as $T \to \infty$, $\bar{\alpha}_T \to 0$, and $\mathbf{x}_T \in \mathcal{N}(\mathbf{0}, \mathbf{I})$. An appealing property of the forward diffusion process is that the reverse conditional probability, when conditioned on $\mathbf{x}_0$, follows a Gaussian distribution:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t\mathbf{I}) \tag{5}$$

where:

$$\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t\right), \tag{6}$$

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \cdot \beta_t \tag{7}$$

Furthermore, the reverse process, as depicted in Figure 4a, also approximates a Gaussian distribution when β_t is sufficiently small. As such, we can train a model p_θ to model the reverse process conditioned on $\mathbf{g}$ and $\mathbf{w}$:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{g}) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, \mathbf{g}), 6_\theta(\mathbf{x}_t, \mathbf{g})) \tag{8}$$

Following the methodology outlined in [12] and [27], we re-parameterize $\mu_\theta(\mathbf{x}_t, \mathbf{g})$ and $6_\theta(\mathbf{x}_t, \mathbf{g})$ as:

$$\begin{aligned} \mu_\theta(\mathbf{x}_t, \mathbf{g}) &= \frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}t}}\epsilon_\theta(\mathbf{x}_t, \mathbf{g})\right) \\ 6_\theta(\mathbf{x}_t, \mathbf{g}) &= \exp\left(\nu_\theta(\mathbf{x}_t, \mathbf{g})\log\beta_t + (1-\nu_\theta(\mathbf{x}_t, \mathbf{g}))\log\tilde{\beta}_t\right) \end{aligned} \tag{9}$$

A neural network is trained to estimate $\epsilon_\theta(\mathbf{x}_t, \mathbf{g})$ and $6_\theta(\mathbf{x}_t, \mathbf{g})$. We employ the same hybrid objective function as detailed in [27]. After the reverse process has been learned, and conditioned on $\mathbf{g}$, we are equipped to generate samples $\mathbf{x}_0$ according to Equation (8), commencing with Gaussian noise $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

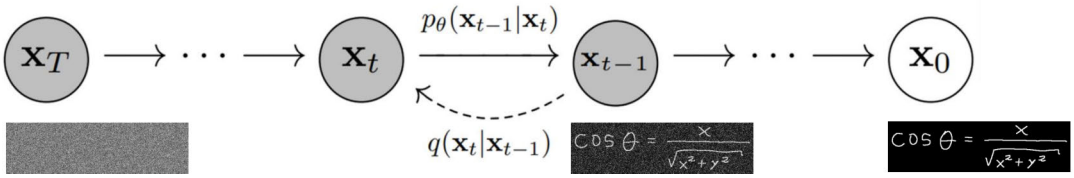
For our task, we employ the same U-Net architecture as in GC-DDPM [9], [12]. As depicted in Figure 4b, both $\mathbf{x}_t$ and $\mathbf{g}$ are standardized to a fixed size. Subsequently, they are concatenated and used as input to the U-Net. The time step t is enriched with a sinusoidal embedding and further processed with a 2-layer FFN. These outputs are summed and fed into the U-Net layers using a feature-wise linear modulation (FiLM) operator [31].

In DDPM, the utilization of classifier-free guidance [16] has proven to be an effective strategy for enhancing generation quality. Following the approach outlined in [12], a content guidance scale γ is introduced. During sampling, $\epsilon_\theta(\mathbf{x}_t, \mathbf{g})$ is replaced with:

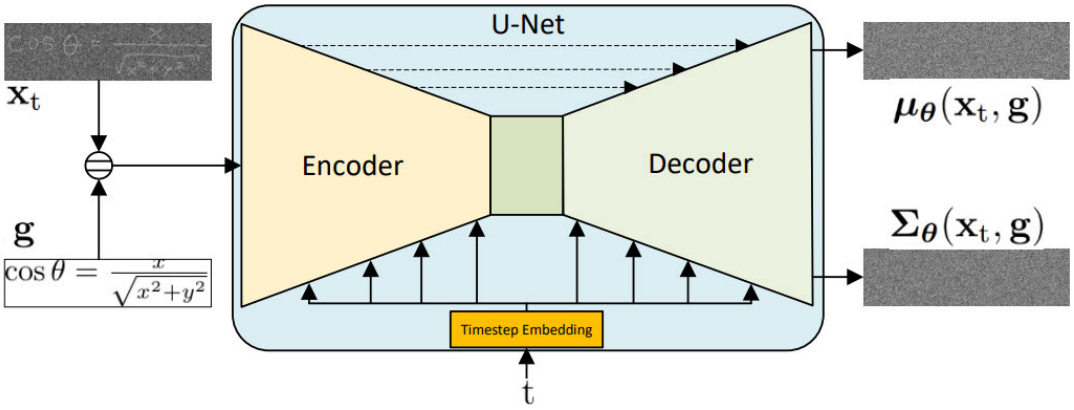
$$\tilde{\epsilon}_\theta(\mathbf{x}_t, \mathbf{g}) = (1+\gamma)\epsilon\theta(\mathbf{x}_t, \mathbf{g}) - \gamma\epsilon_\theta(\mathbf{x}_t\emptyset) \tag{10}$$

Here, $\epsilon_\theta(\mathbf{x}_t, \emptyset)$ is trained concurrently with $\epsilon_\theta(\mathbf{x}_t, \mathbf{g})$ using the same U-Net architecture, with $\mathbf{g}$ being replaced by a special token $\emptyset$ during this process.

In the work presented in [12], the synthesis of offline Chinese character images involved fixed height and width dimensions. However, in the task of generating handwritten mathematical expression images, the dimensions (width and



(a) The Markov chain of forward diffusion process $q(\mathbf{x}_t|\mathbf{x}_{t-1})$ and the learned reverse diffusion process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ of DDPM.



(b) The architecture of the U-Net to estimate $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, \mathbf{g}, \mathbf{w})$ and $\boldsymbol{\Sigma}_\theta(\mathbf{x}_t, \mathbf{g}, \mathbf{w})$. Figures are adapted from [12].

FIGURE 4. Illustration of GC-DDPM framework for handwritten image generation.

height) of the generated images need to be inferred from the LaTeX strings. To address this requirement, we select training images with aspect ratios ranging from 0.125 to 8 in order to construct the diffusion model.

To achieve this, we follow a procedure that includes clockwise rotation of an image if its aspect ratio is less than 1. Subsequently, the maximum edge length is resized to 512 if it exceeds this threshold. Finally, the image is centered and pasted onto a black canvas with a height of 256 and a width of 512. This same procedure is applied to process the glyph images, resizing them to 256×512 dimensions.

IV. EXPERIMENTAL RESULT

In this section we will describe experiments conducted according to the methodology in Section II and III. Section IV-A describes the setting of the comparative experiment and diffusion model training; Section IV-B first comprehensively describes the comparison between the experimental results of the method in this paper and those in other work, and then the influences of some parameters of the three methods in this paper on the results are listed and analyzed to test the sensitivity of the method. In Section IV-C visualization results of diffusion model generated math expression samples are shown and finally we also evaluate the improvements on system accuracy by using these synthetic samples.

A. EXPERIMENTAL SETUP

In our experiments, we utilized the CROHME datasets for both training and testing. Specifically, the CROHME 2014 dataset [25] served as our training set, while the CROHME 2014 test sets were used as the validation set. For testing, we employed the CROHME 2016 [26] and 2019 [24] datasets. The CROHME 2014 dataset comprises a total of 8,836 handwritten mathematical expressions, while the CROHME 2014, 2016, and 2019 test sets contain 986, 147, and 1,199 expressions, respectively. These dataset splits are same with the official setting.

Our implementation is based on the open-sourced BTTR [48]. During training, we refrained from using data augmentation and applied dropout techniques to mitigate overfitting. For the "Tandem" and "Parallel" approaches, the output dimension of the MHSA layers in the Transformer encoder was set to 256, and the hidden dimension of the FFN was set to 1024. In contrast, for the "Mixing" approach, the output dimension of the MHSA layers was set to 192. During decoding, we utilized beam search with a beam size of 10.

The InkML data in the dataset originally contained handwritten brush tracing information and ground truth in LaTeX format. We converted this data into bitmap image format for subsequent training and testing. To ensure uniformity, we applied size normalization to all images by zero-padding, making their widths and heights divisible by 16. Following training and testing, the recognized LaTeX sequence results and ground truth LaTeX texts from the test sets were converted to the symLG format for evaluation using the official evaluation tool.

We evaluated and compared the results of each model using the Expression Recognition Rate (ExpRate), which is the accuracy of correctly recognizing and interpreting the entire mathematical expressions. Additionally, we calculated the ExpRate with different recognition error tolerances of one, two, and three, denoted as '$\leq$ 1', '$\leq$ 2', and '$\leq$ 3',

TABLE 2. Performance comparison of different systems on CROHME 2014/2016/2019 test sets. * denotes that data augmentation is used during training.

Dataset	System	ExpRate(%)	$\leq$ 1(%)	$\leq$ 2(%)	$\leq$ 3(%)	StrRate(%)
CROHME 2014	WYGIWYS [7]	28.7	-	-	-	-
	WAP [46]	46.55	61.16	65.21	66.13	-
	DenseWAP [45]	43	57.8	61.9	-	-
	Pal-v2 [40]	48.88	64.5	69.78	73.83	-
	BTTR [48]	53.96	66.02	70.28	-	-
	SAN [43]	56.20	72.60	79.20	-	-
	ABM [50]	56.04	73.10	79.90	-	-
	CAN [51]	57.26	74.52	82.03	-	-
	SAM-CAN [52]	58.01	-	-	-	-
	ASNRT [53]	57.79	75.01	79.39	-	-
	Li *et al.** [20]	56.59	69.07	75.25	78.60	-
	Ding *et al.** [49]	58.70	-	-	-	-
	SAN* [43]	63.10	75.80	82.00	-	-
	Baseline	55.78	68.56	73.53	75.76	74.04
	Tandem	**57.20**	68.15	73.83	76.37	74.75
	Parallel	56.49	69.27	72.92	74.95	74.24
	Mixing	55.88	68.36	72.92	75.76	74.54
CROHME 2016	WAP [46]	44.55	57.1	61.55	62.34	-
	DenseWAP [45]	40.1	54.3	57.8	-	59.2
	PAL-v2 [40]	49.61	64.08	70.27	73.5	-
	BTTR [48]	52.31	63.9	68.61	-	69.4
	SAN [43]	53.60	69.60	76.80	-	-
	Li *et al.** [20]	54.58	69.31	73.76	76.02	-
	Ding *et al.** [49]	57.72	70.01	76.37	78.90	-
	SAN* [43]	61.50	73.30	81.40	-	-
	ABM [50]	53.36	70.01	78.12	-	-
	CAN [51]	56.15	72.71	80.30	-	-
	SAM-CAN [52]	**56.67**	-	-	-	-
	ASNRT [53]	55.10	70.62	78.12	-	-
	Baseline	53.8	65.5	72.66	75.37	72.4
	Tandem	54.85	67.86	74.50	76.86	74.06
	Parallel	**55.63**	68.82	74.24	76.86	74.76
	Mixing	52.93	66.29	71.88	74.85	73.01
CROHME 2019	DenseWAP [45]	41.7	55.5	59.3	-	60.7
	DenseWAP-TD [45]	51.4	66.1	69.1	-	69.8
	BTTR [48]	52.96	65.97	69.14	-	70.06
	SAN [43]	53.50	69.30	70.10	-	-
	Ding *et al.** [49]	61.38	75.15	80.23	82.65	-
	SAN* [43]	62.10	74.50	81.00	-	-
	ABM [50]	53.71	71.23	78.23	-	-
	CAN [51]	55.96	72.73	80.57	-	-
	SAM-CAN [52]	57.96	-	-	-	-
	ASNRT [53]	56.08	71.48	79.07	-	-
	Baseline	54.14	68.92	74.19	76.36	74.27
	Tandem	**58.56**	72.43	75.69	77.86	76.52
	Parallel	57.39	71.18	75.86	77.94	76.02
	Mixing	55.64	68.76	73.02	75.36	73.52

respectively. We also assessed the structure rates (StrRate), which is the accuracy of correctly recognizing the structure of mathematical expressions, including the relationships and arrangements of symbols (like fractions, exponents, or nested expressions), using the official evaluation tool.

For DDPM training, after filtering images based on aspect ratios, we had 8,474 images as training samples. Our implementation was based on OpenAI's open-sourced guided diffusion [9] and glide-text2im [28]. During training, the glyph condition $\mathbf{g}$ was randomly set to $\emptyset$ with a probability of 10%. We set the batch size to 256, and the image size to 256×512. We used the AdamW optimizer [21] with a learning rate of 1.0e-4.

The U-Net architecture we employed included 2 res-blocks for each stage, and res-blocks were used for downsampling and upsampling feature maps. Channel multiplications for each stage were set to 1, 1, 2, 3, and 4, respectively, and attention resolutions with respect to feature map widths were set at 64, 32, and 16. The diffusion step number was set to 1,000 with a linear noise schedule.

GC-DDPMs were trained for approximately 100,000 steps using a machine equipped with 8 Nvidia A800 GPUs connected by Nvlink, which required approximately 143 hours. During sampling, we utilized the DDPM sampling method with 200 steps to ensure diversity of generated samples. The math expression corpus for generation is same with training set of DDPM model, which is filtered from CROHME 2014 according to aspect ratio and contains 8,474 samples.

B. EXPERIMENTAL RESULTS OF MODEL ARCHITECTURE DESIGN

1) OVERALL RESULTS

In Table 2, we present the results obtained from our proposed baseline, the "Tandem" approach, the "Parallel" approach,

and the "Mixing" approach, alongside comparisons with other AED-based systems. The comparative systems include WYGIWYS, WAP, DenseWAP(-TD), PAL-v2, BTTR, SAN, Li et al., Ding et al, ABM, CAN, SAM-CAN and ASNRT.

Our baseline outperforms BTTR slightly, which can be attributed to the replacement of ReLU with GELU and the adoption of PreNorm instead of PostNorm in all Transformer layers.

To optimize the performance of the proposed approaches, we experimented with varying the number of Transformer encoder layers in the "Tandem" and "Parallel" models, as well as the number of attention heads in the MHSA layers for the "Mixing" models. These variations were determined based on the validation set (CROHME 2014) ExpRates.

Our experimental findings indicate that the best-performing "Tandem" model comprises 12 Transformer encoder layers, while the optimal "Parallel" model incorporates 8 layers. Furthermore, the most effective "Mixing" model employs 12 attention heads in each MHSA layer.

The results reveal that the "Tandem" method performed exceptionally well on the CROHME 2014 dataset, serving as the validation set. On the CROHME 2016 test set, the "Parallel" method achieved the highest ExpRate, reaching 55.63%. Conversely, on the CROHME 2019 test set, there was little disparity in accuracy among the three methods, with the "Tandem" approach having a slight edge. Across all three test sets, the performance of the three methods consistently surpassed that of other AED-based methods without the use of data augmentation. Notably, the "Tandem" method emerged as the best-performing overall. Without using data augmentation, our tandem solution achieves best recognition accuracy on CROHME-2019 test set.

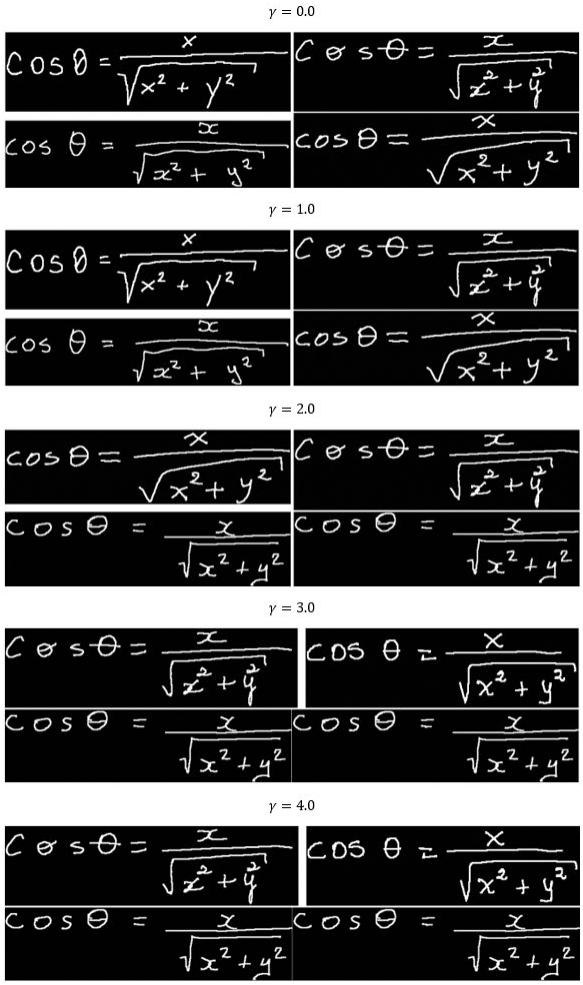


FIGURE 5. Visualization results of different guidance scales.

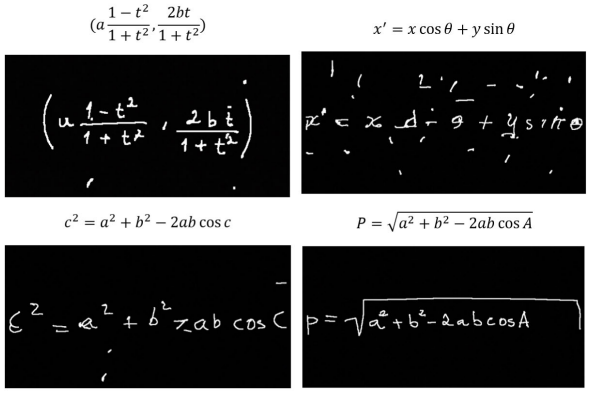


FIGURE 6. Visualization results of some failure synthetic samples.

2) EFFECTS OF NUMBER OF TRANSFORMER ENCODER LAYERS TO TANDEM APPROACH

In this section, we delve into the impact of the number of Transformer encoder layers on the performance of the "Tandem" model. We conducted experiments with varying numbers of layers, specifically 4, 8, 12, and 16 layers. Each layer consists of a MHSA mechanism with 256 output neurons, accompanied by a FFN with 1024 neurons.

The results, as summarized in Table 3, highlight the performance variations across different datasets. Notably, the "Tandem" model with 12 layers achieved the highest ExpRates of 57.20% and 58.56% on the CROHME 2014 and CROHME 2019 datasets, respectively. These results marked an improvement of 1.93% and 0.92% compared to the "Tandem" model with 4 layers.

Increasing the number of layers in the "Tandem" model to around 12 resulted in enhanced performance. However, it's worth noting that excessively increasing the number of layers can overly complexify the model, leading to reduced efficiency and diminishing returns.

3) EFFECTS OF NUMBER OF TRANSFORMER ENCODER LAYERS TO PARALLEL APPROACH

In a similar vein, we conducted an analysis of the impact of the number of Transformer encoder layers on the performance of the "Parallel" approach. We explored different layer configurations, specifically setting the encoder layer count to 4, 8, 12, and 16. It's important to note that the MHSA

TABLE 3. Performance comparison of *Tandem* approach with different number transformer layers on CROHME 2014/2016/2019 test sets.

Dataset	# Layers	ExpRate(%)	$\leq$ 1(%)	$\leq$ 2(%)	$\leq$ 3(%)	StrRate(%)
CROHME 2014	4	55.27	69.47	73.53	76.37	74.54
	8	55.38	67.14	72.82	75.66	74.14
	12	**57.20**	**68.15**	**73.83**	**76.37**	**74.75**
	16	56.49	68.15	72.72	75.56	74.04
CROHME 2016	**4**	**55.55**	**67.86**	**73.54**	**76.16**	**73.97**
	8	54.41	68.47	73.89	76.94	74.67
	12	54.85	67.86	74.50	76.86	74.06
	16	55.28	68.12	73.80	76.42	74.85
CROHME 2019	4	57.64	71.68	76.94	78.53	77.36
	8	57.48	71.18	75.52	77.11	75.86
	12	**58.56**	**72.43**	**75.69**	**77.86**	**76.52**
	16	57.98	71.93	76.69	78.45	76.86

TABLE 4. Performance comparison of *Parallel* approach with different number of transformer encoder layers on CROHME 2014/2016/2019 test sets.

Dataset	# layers	ExpRate(%)	$\leq$ 1(%)	$\leq$ 2(%)	$\leq$ 3(%)	StrRate(%)
CROHME 2014	4	56.39	69.07	74.65	77.08	75.46
	8	**56.49**	**69.27**	**72.92**	**74.95**	**74.24**
	12	55.07	67.95	72.41	74.95	73.23
	16	54.87	68.05	73.02	75.35	74.24
CROHME 2016	4	54.93	68.03	73.97	76.94	75.02
	8	**55.63**	**68.82**	**74.24**	**76.86**	**74.76**
	12	53.89	67.07	72.31	74.67	73.19
	16	54.93	67.42	73.01	75.72	73.89
CROHME 2019	4	56.98	71.26	75.61	77.44	76.19
	8	**57.39**	**71.18**	**75.86**	**77.94**	**76.02**
	12	55.56	68.92	73.35	76.02	74.27
	16	56.73	72.60	76.94	78.70	77.86

TABLE 5. Performance comparison of *Mixing* approach with different number of attention heads on CROHME 2014/2016/2019 test sets.

Dataset	#heads	ExpRate(%)	$\leq$ 1(%)	$\leq$ 2(%)	$\leq$ 3(%)	StrRate(%)
CROHME 2014	1	54.67	67.14	72.11	74.04	72.52
	2	54.46	66.53	71.40	73.94	72.11
	4	54.77	67.75	72.62	75.15	73.33
	6	54.16	66.02	71.10	73.63	71.20
	8	55.38	68.36	74.04	76.17	74.44
	12	**55.88**	**68.36**	**72.92**	**75.76**	**74.54**
CROHME 2016	1	53.28	67.07	72.49	75.46	73.89
	2	54.85	68.03	74.24	76.68	74.93
	4	55.28	67.51	72.31	75.28	73.80
	6	53.54	66.64	72.40	75.20	73.01
	8	**55.46**	**69.17**	**74.59**	**77.55**	**75.37**
	12	52.93	66.29	71.88	74.85	73.01
CROHME 2019	1	56.81	70.68	75.52	77.19	76.19
	2	56.06	69.42	72.85	74.94	73.52
	4	54.05	68.09	72.18	74.44	73.27
	6	56.47	68.92	73.77	76.19	73.93
	8	**57.81**	**71.60**	**75.94**	**77.94**	**77.03**
	12	55.64	68.76	73.02	75.36	73.52

output dimension and FFN hidden dimension in each layer remained consistent with the ''Tandem'' approach.

The results, summarized in Table 4, reveal that across all three test sets, the ''Parallel'' model with 8 Transformer encoder layers achieved the highest ExpRate. This optimal configuration consistently outperformed other layer counts in terms of recognition accuracy.''

4) EFFECTS OF NUMBER OF ATTENTION HEADS TO MIXING APPROACH

In the ''Mixing'' approach, the 3×3 convolutional layers of the last DenseNet stage are substituted with MHSA layers. To assess the impact of the number of attention heads on the final Experiment Recognition Rate (ExpRate), we conducted experiments with varying head counts: 1, 2, 4, 6, 8, and 12, while maintaining a fixed final output dimension of 192.

As summarized in Table 5, our findings indicate that the optimal configuration for validation ExpRate was achieved with 8 attention heads. Remarkably, this 8-head ''Mixing'' model exhibited stability and reached peak ExpRates of 55.46% and 57.81% on the CROHME 2016 and 2019 test sets, respectively. Notably, when compared to the single-head model, the 8-head model demonstrated improvements in ExpRate, with gains of 2.18% and 1% on the CROHME 2016 and 2019 datasets, respectively.

TABLE 6. Performance comparison of using DDPM-based data augmentation on different systems.

Dataset	System	Data Aug.	ExpRate(%)	≤ 1(%)	≤ 2(%)	≤ 3(%)	StrRate(%)
CROHME 2014	Baseline		55.78	68.56	73.53	75.76	74.04
		✓	57.71	69.57	76.27	77.69	75.66
	Tandem		57.20	68.15	73.83	76.37	74.75
		✓	**58.32**	71.30	75.46	78.80	76.57
	Parallel		56.49	69.27	72.92	74.95	74.24
		✓	57.81	71.20	74.85	76.06	75.96
	Mixing		55.88	68.36	72.92	75.76	74.54
		✓	56.80	69.57	74.65	77.08	76.88
CROHME 2016	Baseline		53.8	65.5	72.66	75.37	72.4
		✓	55.78	67.35	75.51	76.87	75.51
	Tandem		54.85	67.86	74.50	76.86	74.06
		✓	57.14	70.07	77.55	80.27	76.87
	Parallel		55.63	68.82	74.24	76.86	74.76
		✓	**58.50**	72.11	78.23	79.59	77.55
	Mixing		52.93	66.29	71.88	74.85	73.01
		✓	55.10	68.71	75.51	76.87	74.15
CROHME 2019	Baseline		54.14	68.92	74.19	76.36	74.27
		✓	56.71	71.81	76.98	77.73	75.48
	Tandem		58.56	72.43	75.69	77.86	76.52
		✓	59.63	75.48	77.98	79.48	78.07
	Parallel		57.39	71.18	75.86	77.94	76.02
		✓	**60.05**	73.56	78.90	81.57	79.07
	Mixing		55.64	68.76	73.02	75.36	73.52
		✓	56.38	70.06	76.23	78.57	75.48

TABLE 7. Effect of amounts of synthetic images on model performance with parallel approach.

Dataset	# Synthetic Images	ExpRate(%)	≤ 1(%)	≤ 2(%)	≤ 3(%)	StrRate(%)
CROHME 2014	/	56.49	69.27	72.92	74.95	74.24
	1×	57.20	70.59	74.04	75.76	75.15
	2×	57.61	71.20	74.75	75.96	75.56
	4×	57.81	71.20	74.85	76.06	75.96
CROHME 2016	/	55.63	68.82	74.24	76.86	74.76
	1×	57.14	69.39	74.83	76.87	75.51
	2×	57.82	72.11	76.19	79.59	77.55
	4×	58.50	72.11	78.23	79.59	77.55
CROHME 2019	/	57.39	71.18	75.86	77.94	76.02
	1×	59.55	72.73	77.73	80.73	78.07
	2×	59.88	73.31	78.57	81.32	78.73
	4×	60.05	73.56	78.90	81.57	79.07

C. EXPRIMENTAL RESULTS OF DDPM-BASED DATA AUGMENTATION

1) VISUALIZATION RESULTS OF DDPM GENERATED SAMPLES

As demonstrated in a previous study [16], the classifier guidance scale plays a pivotal role in achieving a balance between quality and diversity in the generated samples. To assess the behavior of various content guidance scale values (γ), we systematically explore different γ values and generate samples for examination.

To identify the optimal sampling strategy, we initiate our assessment by visualizing the generated results, as illustrated in Figure 5. Large guidance scales result to more printed styles at the sacrifice of diversity.

We also listed some failure synthetic samples in 6. The typical synthetic failures are unexpected dots, short lines and symbol glyph errors. These bad examples are caused by failed alignment between input printed images and generated handwritten images. We leave the alleviation of these phenomena as our future work.

2) EFFECTS OF DDPM GENERATED SAMPLES ON MODEL PERFORMANCE

Based on our visualization results, it is evident that when $\gamma = 0.0$, the synthetic samples exhibit the highest level of diversity. In essence, utilizing a γ value of 0.0 for sampling yields the most diverse synthetic dataset. This diversity is particularly advantageous for enhancing the generalization capabilities of the HMER system.

To leverage this insight, we employ glyph images rendered from the filtered DDPM training set transcripts as input condition images. Using $\gamma = 0.0$, we generate synthetic images for each mathematical expression transcription. To find the best synthetic-real image ratio, we run the generation procedure for 4 times, and mix $1\times$, $2\times$ and $4\times$ synthetic images with real images to build the parallel system. Table 7 shows the effect of synthtic image ratios on final recognition accuracy. Most gains are got after mixing $1\times$ synthetic data with real images, and the performance improvments become marginal after $4\times$ synthetic data are used. So we combine $4\times$ synthetic images with the real

training dataset to re-train the baseline, tandem, and mixing HMER systems.

The ultimate results of this approach are presented in Table 6. The systems under evaluation encompass the Baseline, Tandem, Parallel, and Mixing approaches, both with and without DDPM-based data augmentation. From the outcomes, it becomes evident that DDPM-synthesized samples consistently enhance the ExpRates of all systems. Notably, with data augmentation, the Parallel approach demonstrates a remarkable improvement in ExpRates by 4.6% and 4.4% in relative terms on the CROHME 2016 and 2019 datasets, respectively. These findings substantiate the efficacy of DDPM-synthesized samples in the realm of HMER tasks.

V. CONCLUSION

In this paper, we have explored and proposed innovative approaches for enhancing Handwritten Mathematical Expression Recognition (HMER) systems. We achieved this goal by exploring advanced model architecture design and DDPM-based data augmentation.

Our investigations firstly revolved around three key strategies: the Tandem, Parallel, and Mixing approaches, each designed to leverage the strengths of CNNs and Transformers.

We conducted extensive experiments using the CROHME dataset, evaluating these approaches' effectiveness and their synergy with DDPM-based data augmentation. Our findings revealed that the Tandem approach excels on the CROHME 2014 dataset, while the Parallel approach shines on the CROHME 2016 dataset. Meanwhile, the Mixing approach proves versatile across all datasets.

Moreover, we demonstrated the impact of DDPM-based data augmentation on improving recognition accuracy and structural fidelity. This augmentation method consistently enhanced the performance of all systems, underscoring its potential for widespread application in HMER.

Overall, our research contributes to the evolving field of HMER by introducing novel approaches and highlighting the crucial role of data augmentation in achieving superior recognition results. We anticipate that our findings will inspire further advancements in HMER and foster the development of more accurate and robust systems for HMER.

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